INDR 252 Applied Statistics Lab IV

Understanding Central Limit Theorem and Confidence Intervals

Simulation and Central Limit Theorem

Let X be an exponential random variable with parameter ¼. Then E[X]=4, Var(X)=16, q1=4 ln(4/3)=1.49, q2=median=4ln(2)=2.614, q3=4ln(4)=6.833, and the skewness of X is 2.

1) Generate 1000 exponential random variables with parameter ¼ and store it in Exponential[100,10], so that the matrix has 10 columns and 100 rows.
   a. For each row, take the average of 10 entries and enter this value to the 11th column.
   b. Display the following descriptive statistics for all columns: sample mean, sample standard deviation, minimum, maximum, 1st and 3rd quartiles, median, and skewness. How do these statistics compare with each other as well as E[X] and the standard deviation of X?
   c. Generate the following charts for the 1st and 11th columns: histogram, box plot, probability plot. Compare the shape of these plots with those of exponential and normal random variables. Do you think the averages of 10 exponential random variables look like a normal random variable? Why or why not? Explain.
2) In part (1), you generated 100 samples, each having a sample size of 10. After the lab session, generate 100 samples with varying sample sizes such as 5, 20, 30 and 40. Compute the sample means for each sample, and compare the shapes of the plots for all sample sizes. Can you see how the plots start depicting more and more symmetric distribution as the sample size increases, showing a clear convergence to a normal distribution?

Confidence Intervals

1. Let *X* be a normal random variable with mean μ=5 and σ=1. Generate 100 samples of *n*=10 from the random variable *X*.
2. Compute the sample mean of each sample.
3. Assume that the standard deviation of *X*, σ=1, is known to you. Compute the confidence intervals of μ for each sample at the confidence levels of 1-α=0.95.
4. Plot the sample means and confidence intervals of μ for all samples.
5. Write the necessary formulas to count the number of intervals that contain the population mean μ=5. Repeat the experiment to observe the changes in the confidence intervals and the number of intervals that contain the population mean μ=5.
6. How many of the intervals contain the population mean μ=5? Interpret your experiment by relating your observations with the definition of confidence intervals.
7. Change the confidence level 1-α, take α*=0.01, 0.1*. What is the effect of α on the width of confidence intervals? Explain. How does the number of the intervals that contain the population mean μ=5 change by α*=0.01, 0.05, 0.1*? Why? Repeat the experiment if necessary, to observe the changes in the confidence intervals. Interpret your experiment by relating your observations with the definition of confidence intervals.
8. Change the number of observations in the sample, *n*, take *n*=2, 30. How many of the intervals contain the population mean at α=0.05? Does it change by *n*? Why or why not? Explain. What is the effect of *n* on the width of confidence intervals? Explain.